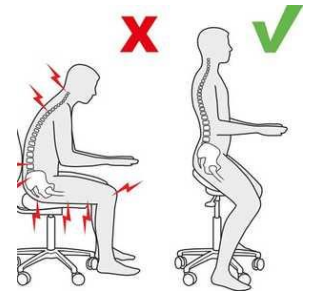


ERGONOMICS IN DENTISTRY

Learning outcomes

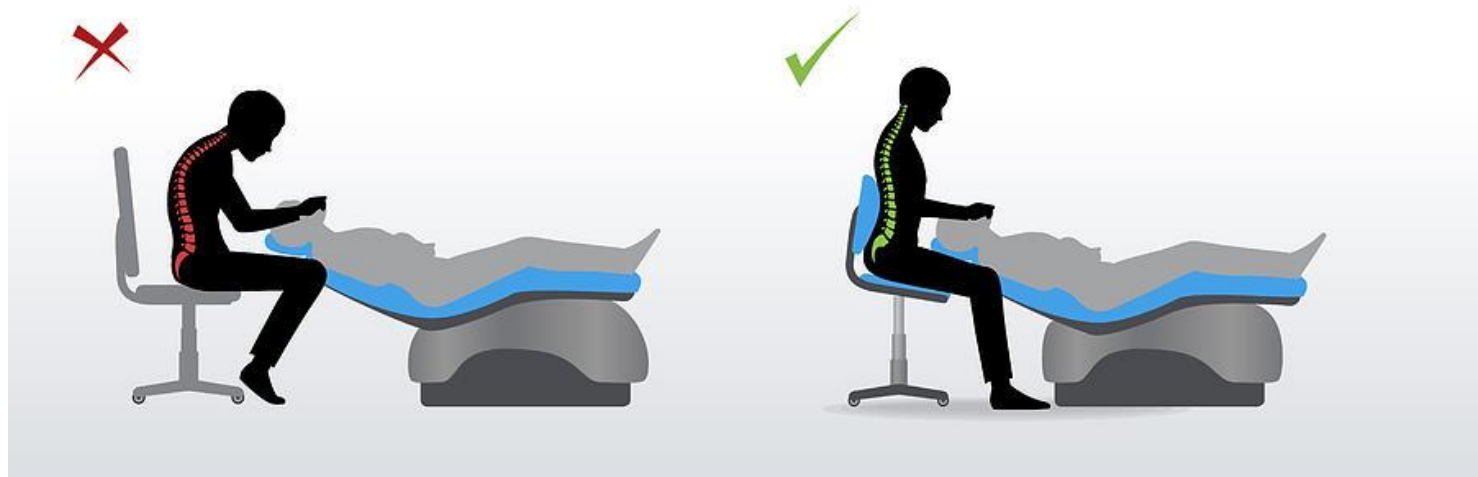
- Recall the best operator position for working on different quadrants
- Explain the instrument exchange zone and four handed dentistry

- The practice of dentistry is demanding and stressful.
- Physical problems may arise if appropriate operating positions are neglected.
- Most restorative dental procedures can be accomplished while seated.



- Positions that create unnecessary curvature of the spine or slumping of the shoulders should be avoided.

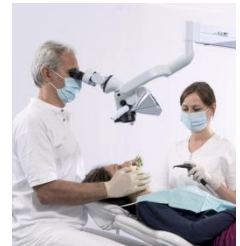
BAD AND PROPER ERGONOMIC POSITIONING





In preparation for a clinical procedure, it is important to ensure that

- Patient and operator(dentist) positions are properly selected, instrument exchange between the dentist and assistant is efficient, and that magnification is used if needed.
- A patient who is in a comfortable position is more relaxed, has less muscular tension, and is more capable of cooperating with the dentist.

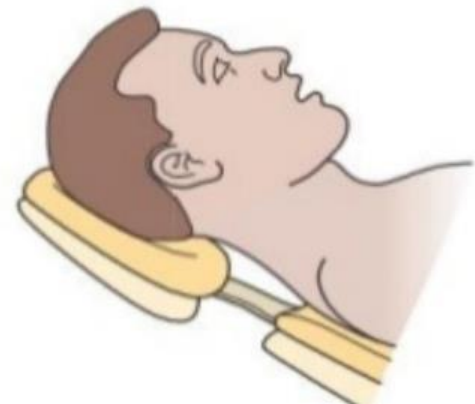
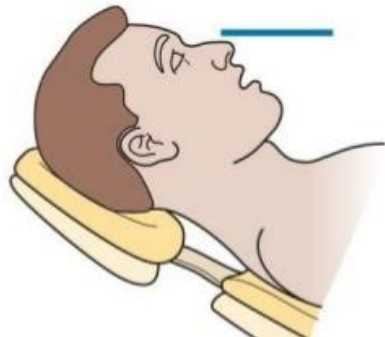


GENERAL CONSIDERATIONS in OPERATIVE DENTISTRY

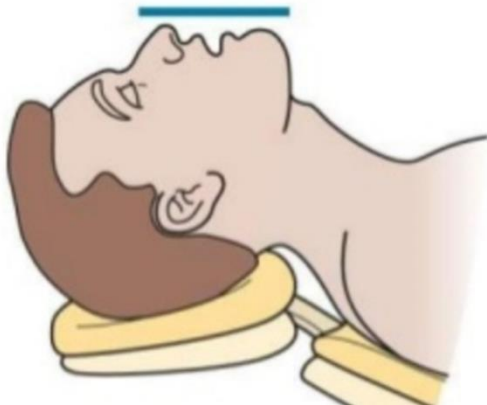
- When operating on mandibular arch, the mandibular occlusal surfaces should be oriented approximately 45 degree with the floor. (chin down)

Patient Head Tilt for Mandibular Arch (Chin-down position)

-Angle the headrest forward and down so that the chin is lower than the nose level



- When operating on the maxillary arch, the maxillary occlusal surfaces should be oriented approximately 90 degree with the floor. (chin up)



- The operator may rotate the patient's head from side to side or from backward or forward, to facilitate visibility and accessibility of the operating field for convenience.



- The face of the operator should not come in close proximity to that of the patient. The ideal distance, similar to that for reading a book, should be maintained
- Another important aspect of proper operating position is to minimize body contact with the patient.
- A proper operator does not rest forearms on the patient's shoulders or hands on the patient's face or forehead.

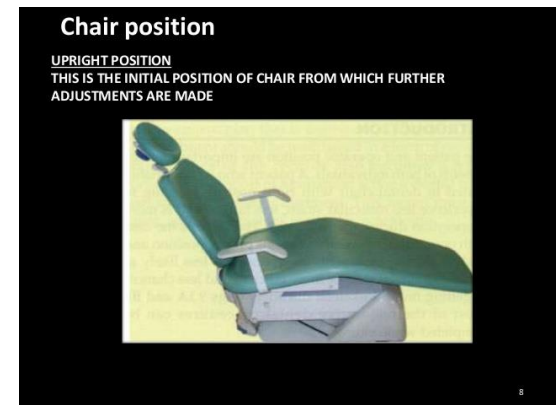
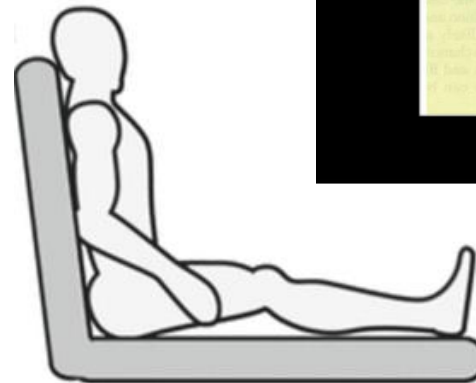


- Unnecessary contact is unpleasant to many patients and should not be practiced.
- From most positions, the left hand should be free to hold the mouth mirror to reflect light onto the operating field to view the cavity preparation indirectly or to retract the cheek or tongue.
- When operating for an extended period, a certain amount of rest and muscle relaxation can be obtained for the operator by changing operating positions reduces muscle strain and fatigue



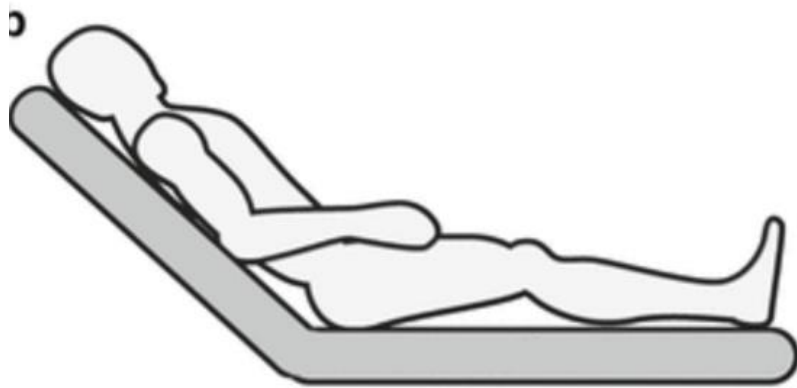
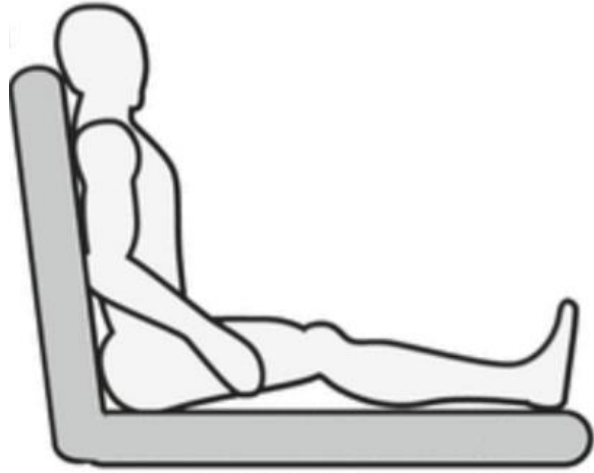
Patient Chair position

- The patient should have direct access to the chair. The chair height should be low, the backrest upright, and the armrest adjusted to allow the patient to get into the chair.



- After the patient is seated, the armrest is returned to its normal position.
- The headrest cushion is positioned to support the head and elevate the chin slightly away from the chest.



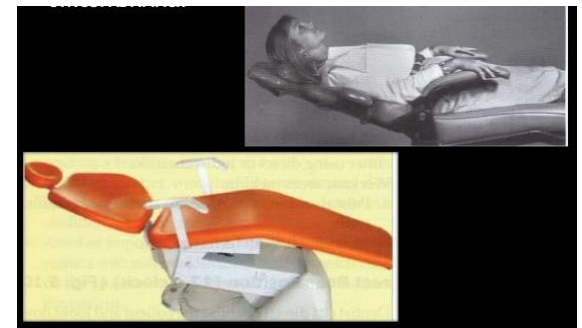
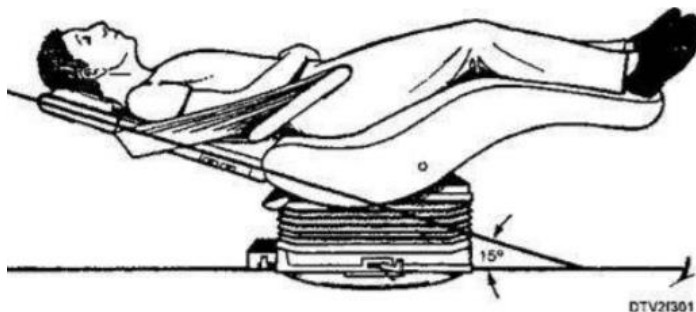


Patient position

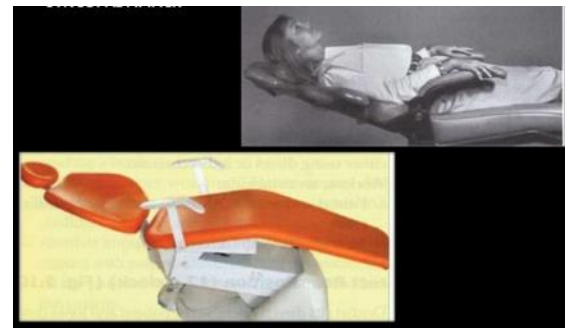
- Reclined 45 degree position



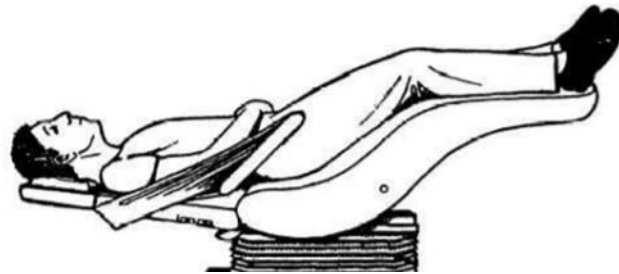
- Supine position, the patient's head, knees and feet are approximately the same level.



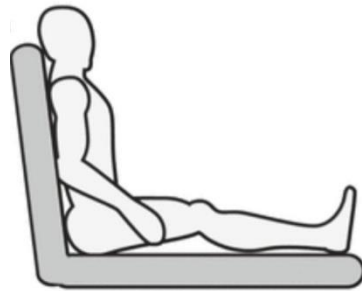
- The choice of patient position varies with the operator, the type of procedure, and the area of the mouth involved in the operation.



- The patient's head should not be lower than the feet; the head should be positioned lower than the feet only in an emergency, as when the patient is in syncope.



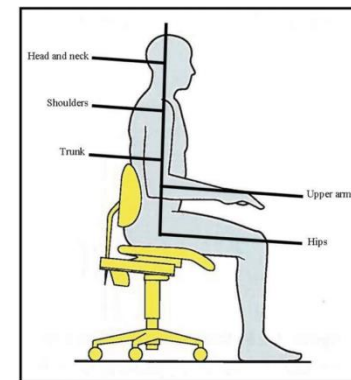
- When the operation is completed, the chair should be placed in the upright position so that the patient can leave the chair easily.



- To improve infection control, chairs with a foot switch for patient positioning are recommended.

Dentist chair position

- Thighs parallel to the floor and knees are apart.
- Shoulders relaxed
- Patient's mouth at elbow height.
- The hips should sit at an angle of 90°
- The operator's back should be straight.



- The operator should sit back on the cushion, using the entire seat and not just the front edge.



- The seated work position for the dental assistant is essentially the same as for the operator except that the dental assistant chair is 4 to 6 inches higher for maximal visual access



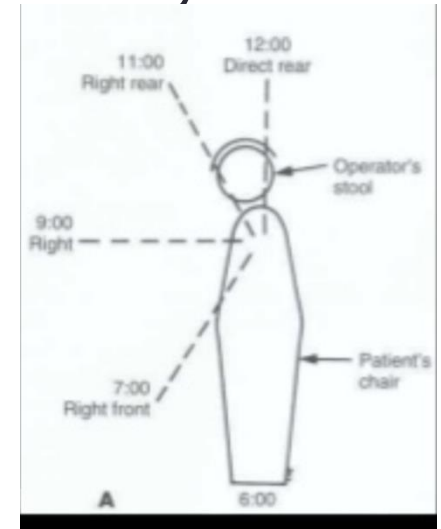
Operating positions

- For a right-handed operator, there are essentially three positions

Right front - 7-o'clock

Direct Right - 9-o'clock

Right rear - 11-o'clock



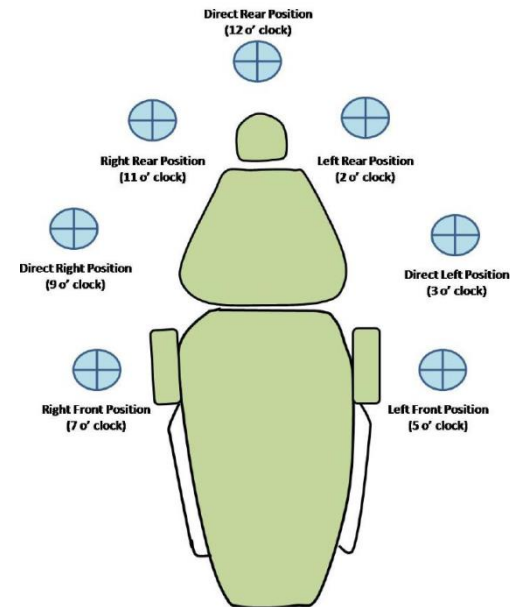
A fourth position, direct rear or 12-o'clock position, has application for certain areas of the mouth.

For a left-handed operator, the three positions are

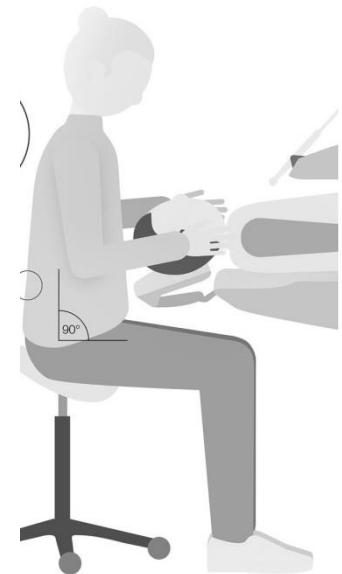
Left front – 5 -o'clock

Direct Left - 3 -o'clock

Left rear – 1 -o'clock positions.



- Most dental treatment is delivered from a seated position.
- As a rule, the teeth being treated should be at the same level as the operator's elbow

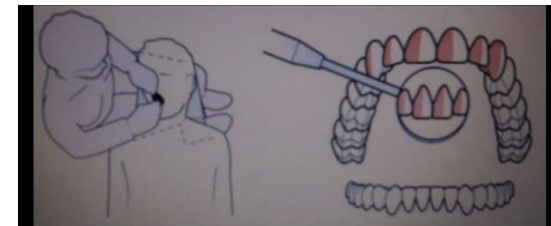
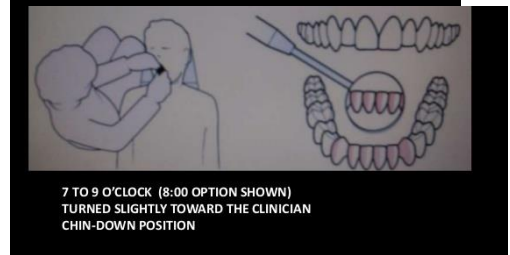


Right Front Position or 7 o'clock position

The right front position facilitates examination and work on

- Facial surface of mandibular anterior teeth
- Facial surface maxillary anterior teeth.

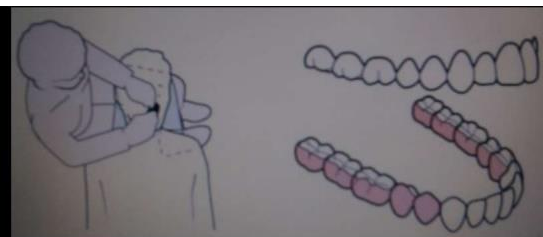
It is often advantageous to have the patient's head rotated slightly toward the operator.



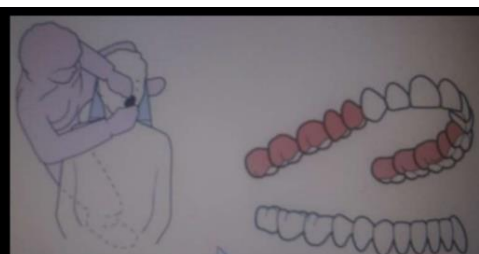
Direct Right Position or 9 o'clock position



- In the right position, the operator is directly to the right of the patient.
- This position is convenient for operating on the facial surfaces of the maxillary and mandibular right posterior teeth and the occlusal surfaces of the mandibular right posterior teeth.



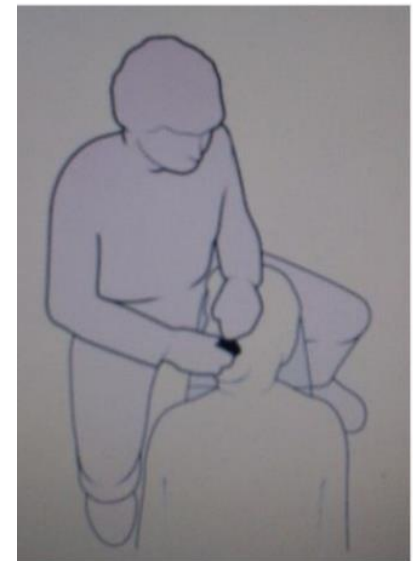
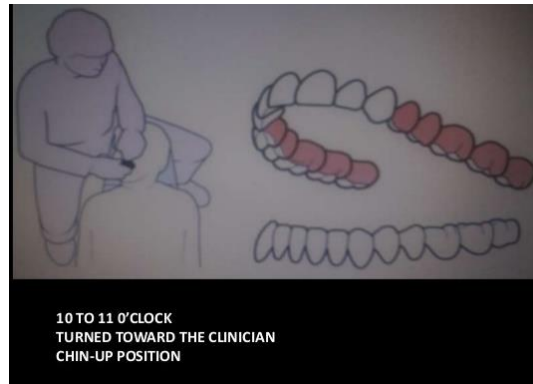
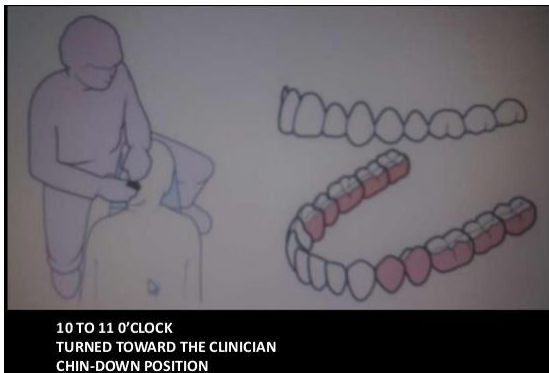
9 O'CLOCK (OPTION 1 FOR 9:00)
TURNED SLIGHTLY AWAY FROM THE CLINICIAN
CHIN-DOWN POSITION



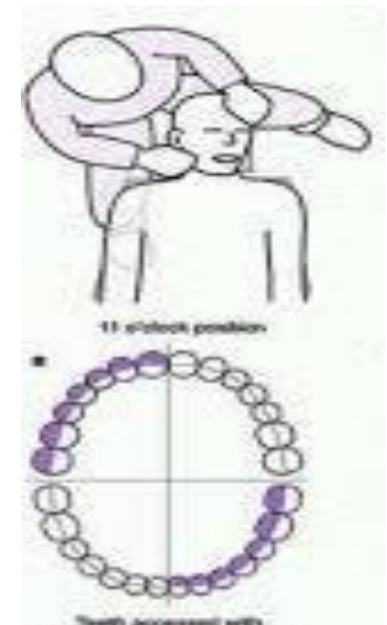
9 O'CLOCK (OPTION 2 FOR 9:00)
TURNED SLIGHTLY AWAY FROM THE CLINICIAN
CHIN-UP POSITION

Right Rear Position

- The right rear position is the position of choice for most operations.
- Most areas of the mouth are accessible and can be viewed directly or indirectly using a mouth mirror.

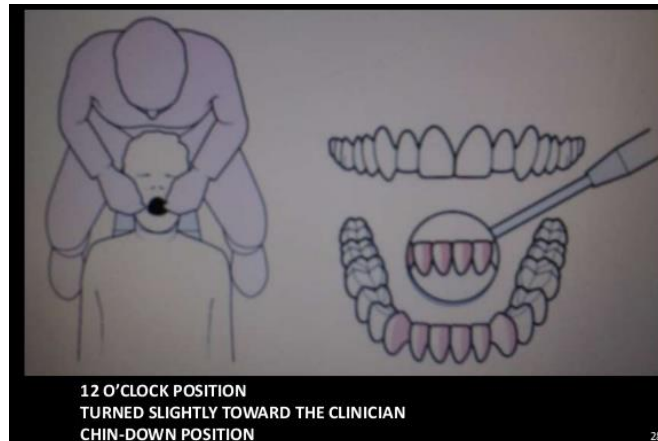


- The operator is behind and slightly to the right of the patient. The left arm is positioned around the patient's head
- When operating from this position, the lingual and incisal (occlusal) surfaces of the maxillary teeth are viewed in the mouth mirror.
- Direct vision may be used on mandibular teeth, particularly on the left side, but the use of a mouth mirror is advocated for visibility, light reflection, and retraction.



Direct Rear Position

- The direct rear position has limited application and is used primarily for operating on the lingual surfaces of mandibular anterior teeth.
- The operator is located directly behind the patient and looks down over the patient's head



Musculoskeletal disorders

- Musculoskeletal disorders MSD'S (29.5%) is a common cause of early retirement for dentists

(Burke et al 1997)

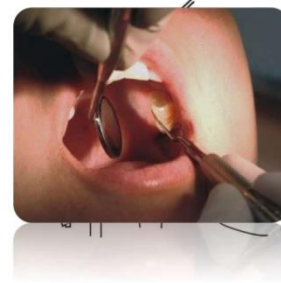
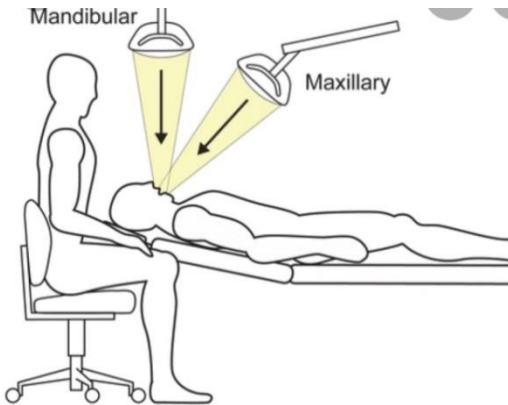
Types of MSD'S occurs in dentists.

1. Lower back pain
2. Upper back pain
3. Wrist Tendonitis(Inflammation of a tendon in the wrist)
4. Carpel tunnel syndrome(Caused by the pressure on the median nerve in the carpal tunnel of wrist), symptoms include numbness, tingling and weakness in the thumb and fingers



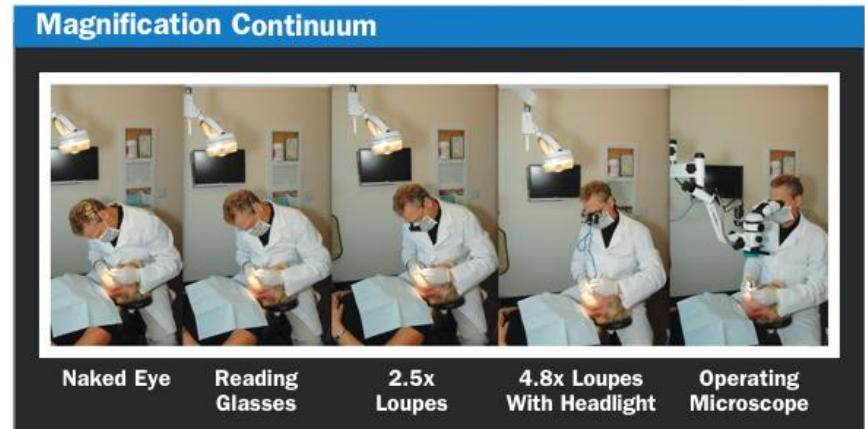
Lighting

- The goal of overhead lighting is to produce even, shadow free illumination that is concentrated on the operating field.
1. Overhead light should be positioned as close as possible to the sight line
 2. Hand mirrors to reflect the light intra orally.
 3. Use fiber optics in handpiece.



Magnification

- Use of loupes or operating microscope
- *Improved ergonomics*
- *Improved visibility of the working area*
- *Reduced eyestrain*



Four Handed Dentistry

“It is an ergonomically sound way to practice dentistry using the skills of the dental assistant while including work simplification techniques.”

Goal: to allow the dentist and assistant to function as a team in a seated position with maximal efficiency and minimal strain.



Zones of activity

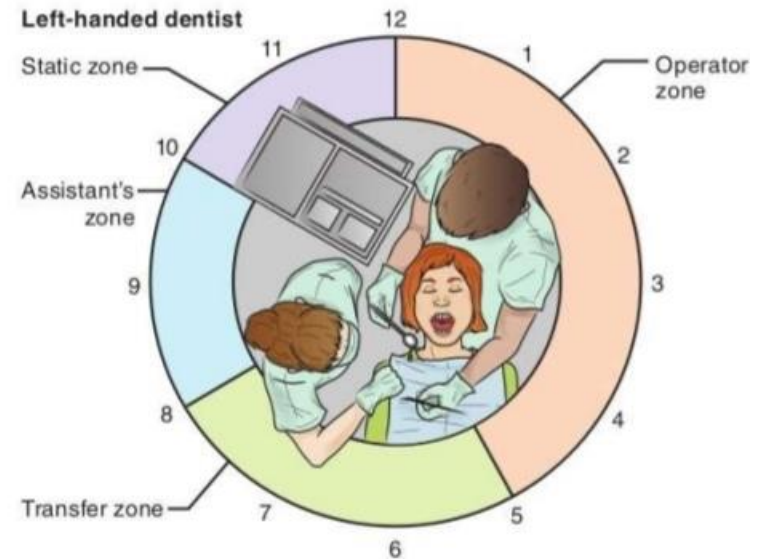
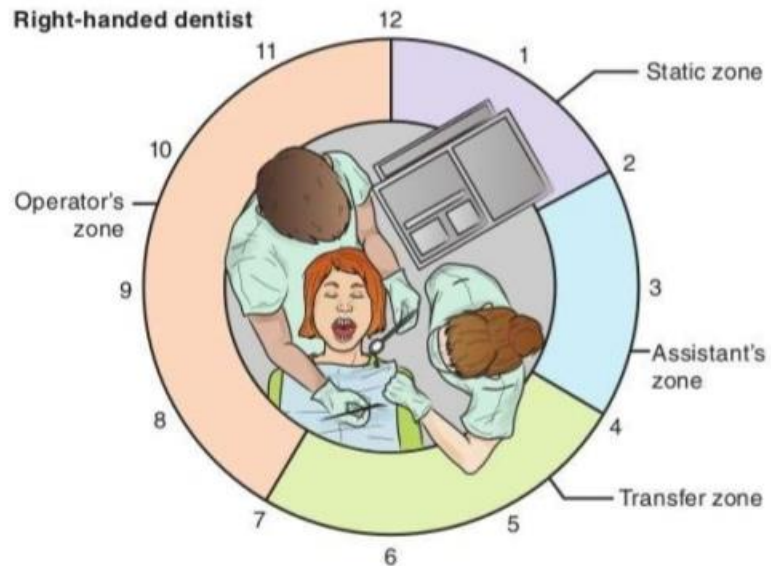
- The work area around the patient is divided into four “zones of activity”.
- Zones of activity are identified using the patient’s face as the face of a clock.
- The four zones are:
 - a) Operator’s zone.
 - b) Assistant’s zone.
 - c) Transfer zone.
 - d) Static zone.

FOUR HANDED DENTISTRY

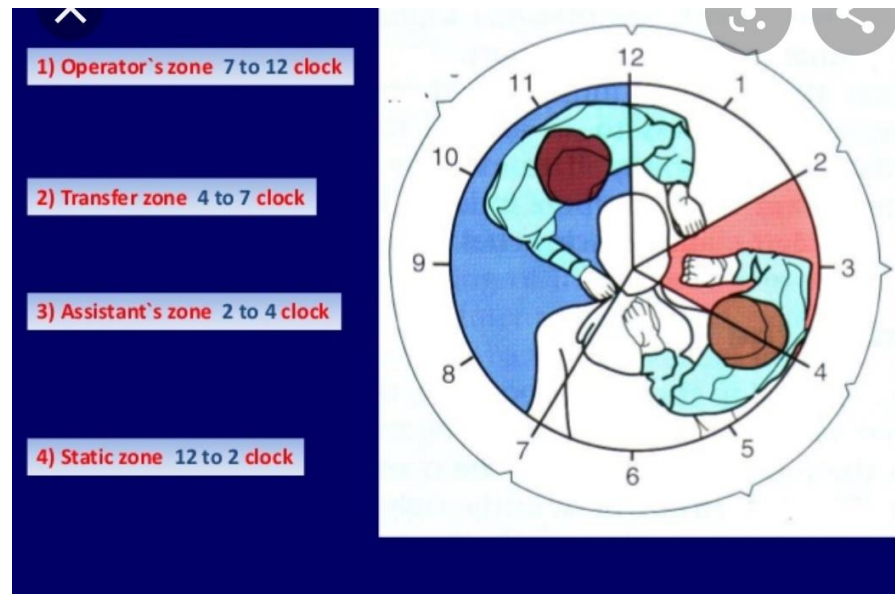


Zones of Activity

Four handed dentistry:



- The static zone, which is the zone of least activity. Instruments that are infrequently used such as the blood pressure equipment and portable curing light can be stored in this area.



- All instrument exchanges between the operator and assistant should occur in the exchange zone below the patient's chin and several inches above the patient's chest.
- Instruments should not be exchanged over the patient's face.

APART FROM ERGONOMICS

- Proper scheduling of patients - sufficient recovery time from muscle fatigue
- Personal protective equipment - proper glasses, clothing, gloves
- Exercise and fitness of operator.

References

- Art and science of operative dentistry sturdevant 5 th edition page no 449 – 452